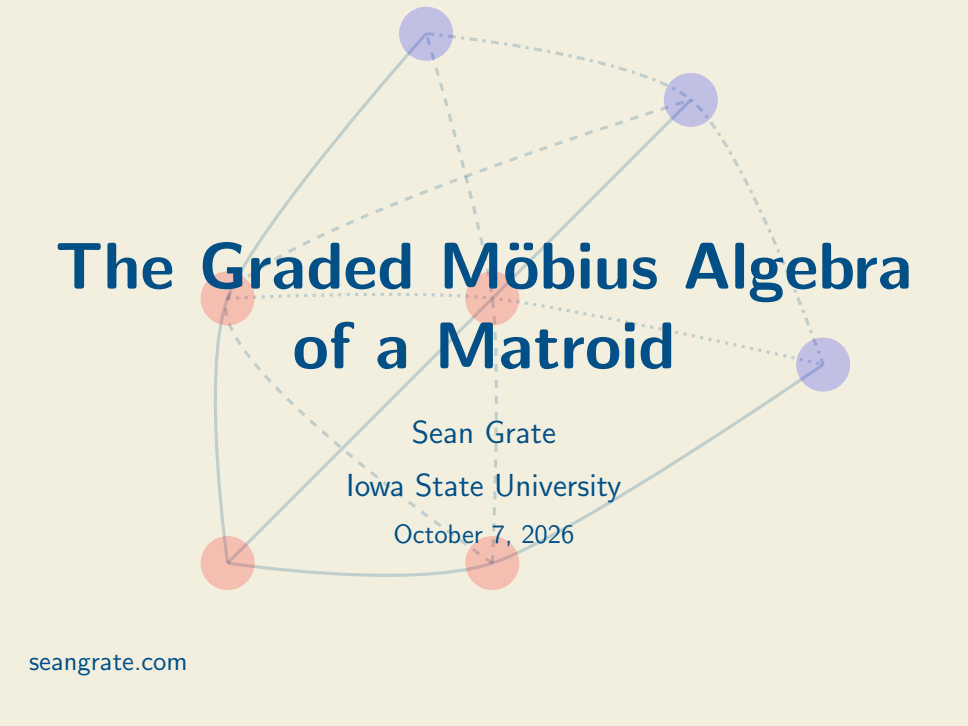




The Graded Möbius Algebra of a Matroid

Sean Grate

Iowa State University

October 7, 2026

Macaulay2 at Auburn 2027



February 22-25th, 2027

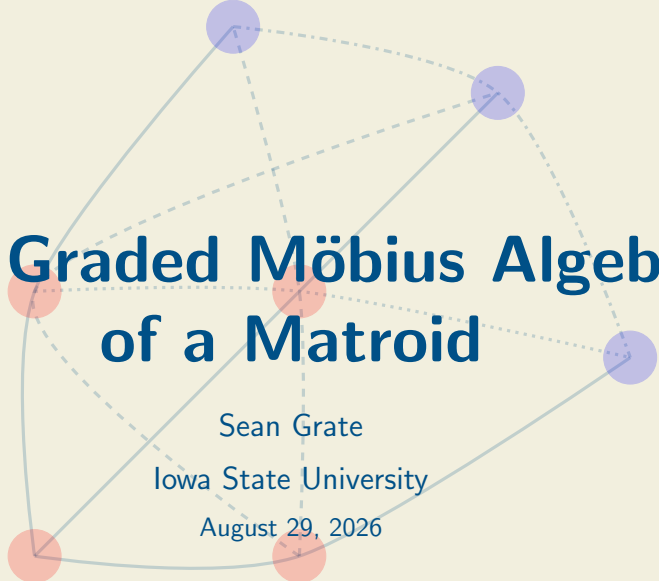
Organizers: John Cobb, Sean Grate, Michael Brown

<https://johndcobb.github.io/organizing/M2atAuburn/>

Objectives

- Develop new software for Macaulay2 and its packages
- Form new collaborations between different mathematical areas
- Train students through hands-on learning and coding

Group Leader	Focus
Keller VandeBogert	<i>Varieties</i> package
Jay Yang	Resolutions of toric subvarieties
Thomas Brazelton	Grothendieck-Witt rings
Jordy Garcia-Lopez	tropical intersections
Sean Grate	<i>Matroids</i> package
Doug Torrance	p -adic numbers
Juliette Bruce & John Cobb	toric stacks



The Graded Möbius Algebra of a Matroid

Sean Grate

Iowa State University

August 29, 2026

The Crew



Sean Grate (Me)



Jason McCullough

Free Resolutions

Definition

Let $I \subseteq S = \mathbb{K}[x_1, \dots, x_n]$ be a homogeneous ideal, and consider the graded minimal free resolution of $R = S/I$:

$$0 \leftarrow R \leftarrow F_0 \leftarrow F_1 \leftarrow \cdots \leftarrow F_{n+1} \leftarrow 0,$$

where $F_i \cong \bigoplus_j S(-i-j)^{\beta_{i,i+j}}$.

- $\beta_{i,i+j}$ are the **graded Betti numbers**
- $\beta_i = \sum_j \beta_{i,i+j}$ are the **total Betti numbers**



Example

$S = \mathbb{K}[x, y, z]$ and $I = \langle x^2, y^2, z^3 \rangle$

$$0 \leftarrow S/I \leftarrow S \xleftarrow{(x^2 \ y^2 \ z^3)} \begin{matrix} S(-2)^2 \\ \oplus \\ S(-3) \end{matrix} \xleftarrow{\begin{pmatrix} -y^2 & -z^3 & 0 \\ x^2 & 0 & -z^3 \\ 0 & x^2 & y^2 \end{pmatrix}} \begin{matrix} S(-4) \\ \oplus \\ S(-5)^2 \end{matrix} \xleftarrow{\begin{pmatrix} z^3 \\ -y^2 \\ x^2 \end{pmatrix}} S(-7) \leftarrow 0$$

	0	1	2	3
0	1	.	.	.
1	.	2	.	.
2	.	1	1	.
3	.	.	2	.
4	.	.	.	1

Example

$$S = \mathbb{K}[x, y, z] \text{ and } I = \langle x^2, y^2, z^3 \rangle$$

$$0 \leftarrow S/I \leftarrow S \leftarrow \begin{matrix} S(-(1+1))^2 \\ \oplus \\ S(-(1+2)) \end{matrix} \leftarrow \begin{matrix} S(-(2+2)) \\ \oplus \\ S(-(2+3))^2 \end{matrix} \leftarrow S(-(3+4)) \leftarrow 0$$

	0	1	2	3
0	1	.	.	.
1	.	2	.	.
2	.	1	1	.
3	.	.	2	.
4	.	.	.	1

Invariants from the Betti Table

	0	1	2	3
0	1	.	.	.
1	.	2	.	.
2	.	1	1	.
3	.	.	2	.
4	.	.	.	1

- Castelnuovo-Mumford regularity = height of the Betti table

$$\operatorname{reg}(S/I) = 4$$

- Projective dimension = length of the Betti table

$$\operatorname{pdim}(S/I) = 3$$

- Hilbert series

$$H_{S/I}(t) = \frac{1}{(1-t)^3} - \left(\frac{2t^2}{(1-t)^3} + \frac{t^3}{(1-t)^3} \right) + \dots = 1 + 3t + 4t^2 + \dots$$

Koszul Algebras

R a (standard-graded) $\mathbb{K}$ -algebra

$$\mathfrak{m} = \bigoplus_{i \geq 1} R_i$$

Definition

R is **Koszul** if $R/\mathfrak{m} \cong \mathbb{K}$ has a linear free resolution over R .

Example

$$R = \mathbb{K}[x, y, z]$$

$$0 \leftarrow R/\mathfrak{m} \leftarrow R \xleftarrow{\begin{pmatrix} x & y & z \end{pmatrix}} R(-1)^3 \xleftarrow{\begin{pmatrix} -y & -z & 0 \\ x & 0 & -z \\ 0 & x & y \end{pmatrix}} R(-2)^3 \xleftarrow{\begin{pmatrix} z \\ -y \\ x \end{pmatrix}} R(-3) \leftarrow 0$$

	0	1	2	3
0	1	3	3	1

More Examples

- Quadratic complete intersections (Tate, 1957)
- Universal enveloping algebras (Priddy, 1970)
- Quotients by quadratic monomial ideals (Fröberg, 1975)
- Sufficiently high Veronese subalgebras of a standard-graded $\mathbb{K}$ -algebra (Backelin, 1986)
- Coordinate rings of Grassmannians (Kempf, '90) and general canonical curves (Vishik-Finkelberg, 1993)

When is a $\mathbb{K}$ -algebra Koszul?

Theorem (Conca-Rossi-Valla, 2001)

Suppose R is a graded Gorenstein $\mathbb{K}$ -algebra presented by quadrics. If $\operatorname{reg}(R) \leq 2$ or if $\operatorname{reg}(R) = 3$ and $\operatorname{codim}(R) \leq 4$, then R is Koszul.

Theorem (Mastroeni-Schenck-Stillman, 2021/2023)

For all integers $c \geq r + 2 \geq 6$, there exists a graded, quadratic, Gorenstein $\mathbb{K}$ -algebra R with $\operatorname{reg}(R) = r$ and $\operatorname{codim}(R) = c$ that is not Koszul.

Theorem (Roos, 1993)

Koszulness of a $\mathbb{K}$ -algebra cannot definitively be checked by computing finitely many steps of the minimal free resolution.

Properties of Koszul Algebras

Definition

The **Poincaré series** $P_R(t)$ of R is the generating function for the total Betti numbers of $R/\mathfrak{m} \cong \mathbb{K}$.

Theorem

If $R = S/I$ is Koszul, then

- *I is generated by quadrics;*
- *$\text{reg}(R) \leq \text{pdim}(R)$;*
- *$\beta_{i,i+j} = 0$ for $j > i$;*
- *$\beta_{i,2i} \leq \binom{c}{i}$, where c is the number of quadrics;*
- *$P_R(t)H_R(-t) = 1$.*

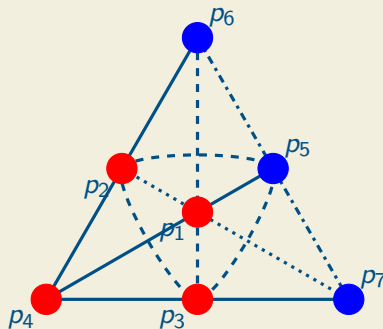
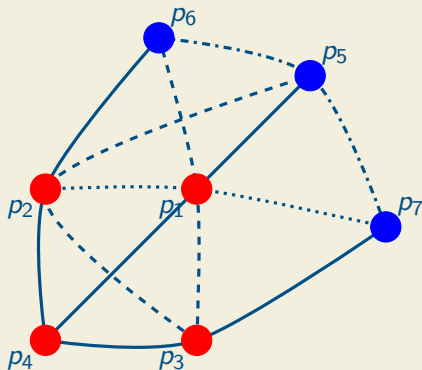
Proving Koszulness

- Construct a Koszul filtration.
 - A family of ideals closed under a certain colon operation (roughly).
 - Typically, the filtration is very long.
- Show R is G -quadratic.

$$\left\{ \begin{array}{l} \text{quadratic Gröbner basis} \\ \text{graded Betti numbers} \\ \text{upper-semicontinuous} \end{array} \right. \implies R \text{ Koszul}$$

Projective Geometries

$$\begin{array}{ccccccc}
 p_1 & p_2 & p_3 & p_4 & p_5 & p_6 & p_7 \\
 \left(\begin{array}{ccccccc}
 1 & 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 1 & 0 & 0 & 1 & 1 & 0 \\
 1 & 0 & 1 & 0 & 1 & 0 & 1
 \end{array} \right)
 \end{array}$$



Matroids Crash Course

Definition

A matroid $M = (E, \mathcal{I})$ consists of a ground set E and independent sets $\mathcal{I} \subseteq 2^E$ such that

- (1) $\emptyset \in \mathcal{I}$;
- (2) If $I' \subseteq I$ and $I \in \mathcal{I}$, then $I' \in \mathcal{I}$;
- (3) If $I, J \in \mathcal{I}$ and $|I| < |J|$, then there exists $e \in J \setminus I$ such that $I \cup \{e\} \in \mathcal{I}$.

Example

- Vector spaces
- Cycles of a graph
- Hyperplane arrangements
- Boolean lattices

Matroids Crash Course

Let $M = (E, \mathcal{I})$ be a matroid.

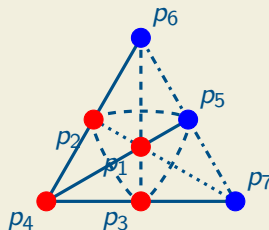
- Elements of $\mathcal{I}$ are called **independent** sets; maximal ones are called **bases**.
- Sets not in $\mathcal{I}$ are called **dependent** sets; minimal ones are called **circuits**.
- The **rank** of a set $F \subseteq E$ is the cardinality of a maximal independent set in F . The **closure** of a set $F \subseteq E$ is

$$\overline{F} := \{e \in E : \text{rank}(F \cup \{e\}) = \text{rank}(F)\}.$$

A set F such that $\overline{F} = F$ is called a **flat**.

$$\mathbf{PG}(2, 2) = F_7$$

$$\begin{matrix} p_1 & p_2 & p_3 & p_4 & p_5 & p_6 & p_7 \\ \left(\begin{array}{cccccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right) \end{matrix}$$



Set	Type	Rank	Closure
$\{p_2, p_3, p_4\}$	basis	3	$\{p_1, p_2, p_3, p_4, p_5, p_6, p_7\}$
$\{p_2, p_3\}$	independent	2	$\{p_2, p_3, p_5\}$
$\{p_1, p_2, p_3, p_4\}$	circuit	3	$\{p_1, p_2, p_3, p_4, p_5, p_6, p_7\}$
$\{p_3, p_4, p_7\}$	dependent, flat	2	$\{p_3, p_4, p_7\}$

Geometric Lattices

Definition

A **lattice** is a poset with well-defined join $\vee$ and meet $\wedge$ operations.

Definition

A lattice is called **geometric** if it is

- (1) atomic
- (2) (upper) semimodular

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) \leq \text{rank}(F) + \text{rank}(G)$$

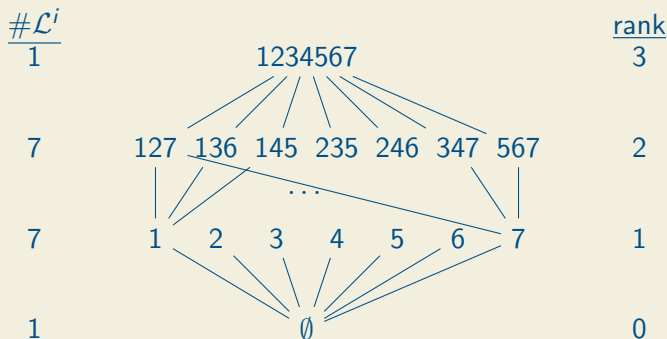
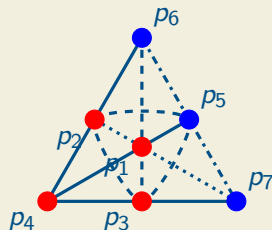
$$\dim(U + V) + \dim(U \cap V) \leq \dim(U) + \dim(V)$$

Theorem

A lattice is geometric if and only if it is the lattice of flats of a matroid.

The Lattice of Flats

$$\begin{matrix}
 p_1 & p_2 & p_3 & p_4 & p_5 & p_6 & p_7 \\
 \begin{pmatrix}
 1 & 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 1 & 0 & 0 & 1 & 1 & 0 \\
 1 & 0 & 1 & 0 & 1 & 0 & 1
 \end{pmatrix}
 \end{matrix}$$



How do different ranks compare?

Theorem (Greene, 1970)

Let $\mathcal{L}$ be a geometric lattice with $\text{rank } \mathcal{L} = n > 1$, and with

$$\mathcal{L}^k := \{F \in \mathcal{L} : \text{rank}(F) = k\},$$

let $W_k = |\mathcal{L}^k|$. Then

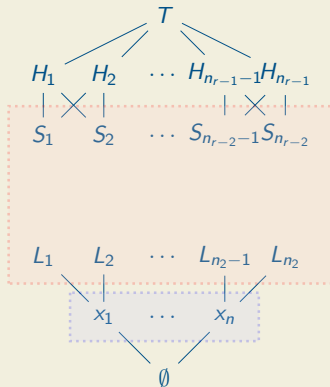
- (1) $W_1 \leq W_k$ for all $k = 2, \dots, n-1$; and
- (2) $W_1 = W_k$ if and only if $k = n-1$ and L is modular.

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) = \text{rank}(F) + \text{rank}(G)$$

The Top-Heavy Conjecture

Theorem (Huh-Wang, 2017; Braden-Huh-Matherne-Proudfoot-Wang, 2020)

Let $\mathcal{L}$ be a geometric lattice of rank r . For any $k \leq j \leq r - k$, there is an injective map $\iota: \mathcal{L}^k \rightarrow \mathcal{L}^j$.



Graded Möbius Algebras

Definition

Let $\mathcal{M}$ be a simple matroid with lattice of flats $\mathcal{L}(M)$. The **graded Möbius algebra** of M is the standard graded $\mathbb{K}$ -algebra

$$B_M = \bigoplus_{F \in \mathcal{L}(M)} \mathbb{K}x_F,$$

with multiplication

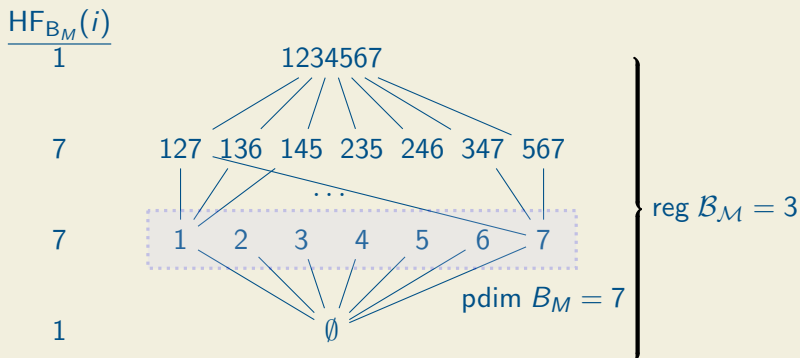
$$x_F x_G = \begin{cases} x_{F \vee G} & \text{if } \text{rank}(F \vee G) = \text{rank}(F) + \text{rank}(G), \\ 0 & \text{otherwise} \end{cases}.$$

Properties

Fact

Let M be a simple matroid with lattice of flats $\mathcal{L}(M)$.

- (1) The Hilbert function of B_M is the rank-generating function of $\mathcal{L}(M)$
- (2) $\text{reg } B_M = \text{rank } M = \text{rank } \mathcal{L}(M)$
- (3) $\text{pdim } B_M = |E(M)|$



When is B_M Koszul?

Theorem (Laclair-Mastroeni-McCullough-Peeva, 2025)

Let $\mathcal{M}$ be a simple matroid. There is a presentation of B_M :

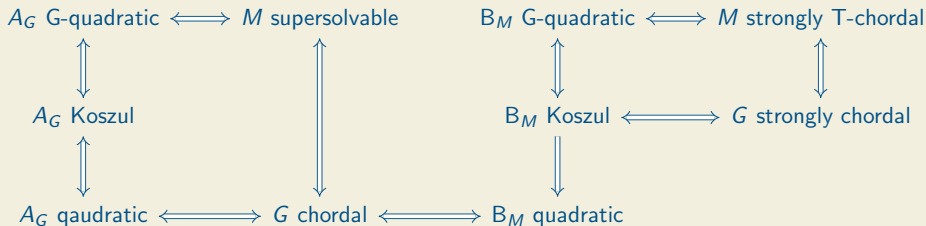
$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{(y_i^2 : i \in E) + (y_{C \setminus i} - y_{C \setminus j} : C \in \mathcal{C}(M))}$$

Theorem (Laclair-Mastroeni-McCullough-Peeva, 2025)

Let M be the cycle matroid of a graph G . Then B_M is Koszul if and only if G is strongly chordal.

Orlik-Solomon + Graded Möbius Algebra Parallels

Let $M = M(G)$ be the cycle matroid of a graph G , and let A_M be its Orlik-Solomon algebra and B_M its graded Möbius algebra.



Projective Geometries

Theorem (Greene, 1970; Maeno-Numata, 2016)

Let M be a simple matroid and $\mathcal{L}(M)$ be its lattice of flats. The following are equivalent:

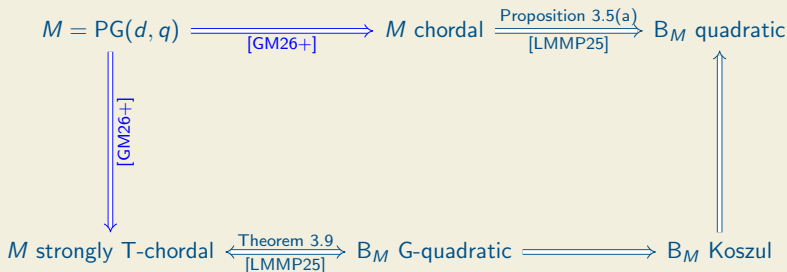
- (1) $\mathcal{L}(M)$ is modular.
- (2) The rank-generating function of $\mathcal{L}(M)$ is symmetric.
- (3) The Hilbert function of B_M is symmetric.
- (4) B_M is Gorenstein.

$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{\text{Ann} \left(\sum_{B \in \mathcal{B}(M)} y_B \right)}$$

Projective Geometries

Theorem (Grate-McCullough, 2026+)

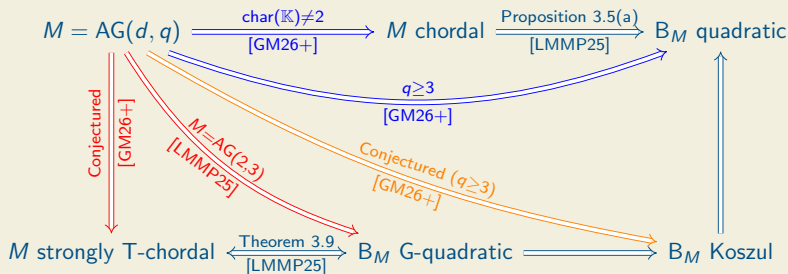
If M is a projective geometry, then B_M is Koszul.



Affine Geometries

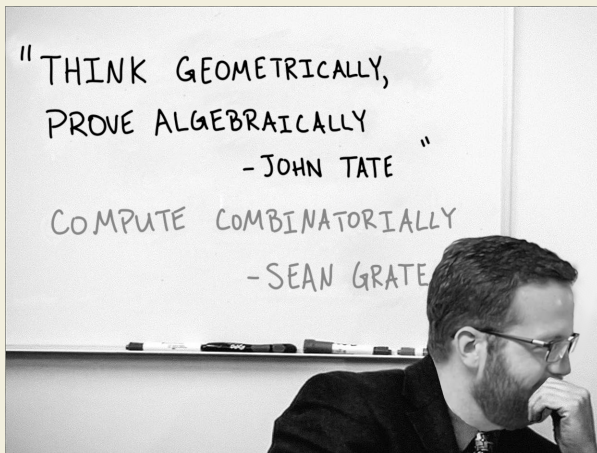
Fact

Affine geometries are obtained from projective geometries by deleting the points on the hyperplane at infinity.



Future Directions

- (1) Can we (combinatorially) compute a graded minimal free resolution of B_M ?



Theorem (FGK+, 2026)

We can compute the Betti numbers for uniform matroids $U_{r,n}$.

Example

The Betti table for $B_{U_{6,8}}$ is

	0	1	2	3	4	5	6	7	8
0	1	.	.	.	.	.	.	.	.
1	.	8	.	.	.	.	.	.	.
2	.	.	28	.	.	.	.	.	.
3	.	.	.	56	.	.	.	.	.
4	.	.	.	.	70	.	.	.	.
5	.	27	208	693	1296	1526	1008	350	48
6	.	.	.	.	.	.	.	.	1

Example

Let $M = \text{PG}(2, 2)$. It is a rank 3 matroid on 7 elements with rank generating function $1 + 7t + 7t^2 + t^3$. The Betti table for B_M is

	0	1	2	3	4	5	6	7
0	1	.	.	.	.	.	.	.
1	.	21	64	70	14	.	.	.
2	.	.	.	14	70	64	21	.
3	.	.	.	.	.	.	.	1

Future Directions

(2) What other matroids have Koszul graded Möbius algebras?

Matroid	B_M Koszul?
$M(G)$	G is strongly chordal
$PG(n, q)$	$n \geq 2$
$AG(n, q)$	$n \geq 2, q \geq 3$ (conjectured)
Sylvester	not in general
Lattice Path	???
Transversal	???
(Sparse) Paving	???
Dowling geometry	???

Thank you!